STEP 1 · CORE-TIER BUILD · NO AI, NO CLOUD

AgriRover

Basic Bot — phone-driven soil rover

A single **ESP32 DevKit V1** runs the whole machine: tank drive, ultrasonic obstacle guard, 7-in-1 soil probe, fertilizer micro-dosing and one-second-per-line telemetry. It hosts its **own WiFi network and control page**, so a farmer drives it from any phone with **no router, no app install and no internet.**

TEAM

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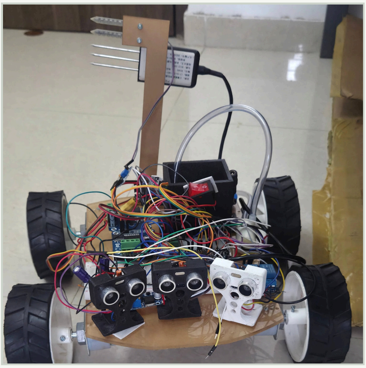
01 WHY BUILD THE BASIC BOT FIRST



As built: 4-wheel deck, 3× HC-SR04 front bumper, servo-raised 7-in-1 probe on the mast, 11.1 V pack, dosing line.

- **Soil testing today is a queue, not a habit.** A grower sends one sample to a lab per season, so fertilizer goes on by convention rather than by reading.
- **The mechanics must be trustworthy before intelligence is added.** Drive, power, grounding, interlocks and dosing have to survive the field first — autonomy on an unreliable base only hides faults.
- **Every part of it must be repairable in a village.** One ₹-cheap ESP32, hobby modules, a phone browser and a USB cable: no Pi, no accelerator, no subscription, no coverage.

IN THIS BUILD

Tank drive · obstacle guard · NPK / moisture / DHT22 / battery · servo probe insertion · micro-dosing · latched halt · phone page · ESP32-CAM video · USB teleop + CSV log

DELIBERATELY DEFERRED

On-board AI (weed / disease), Raspberry Pi link, MQTT & cloud, GPS navigation, encoders, OTA, LoRa. Wiring is pin-compatible with the full firmware, so the upgrade needs no rewiring.

02 ARCHITECTURE — ONE CONTROLLER, PRIVATE NETWORK

CONTROLLER

ESP32 DevKit V1

`FreeRTOS · 2 tasks · 5 s task WDT`

- Drive + command loop @ 50 Hz on core 1
- Sensors @ 5 Hz, telemetry @ 1 Hz on core 0
- 2× BTS7960 tank drive on LEDC PWM
- NPK 7-in-1 over RS485 / Modbus RTU
- Moisture, DHT22, 3S divider, relay pump, MG995 probe

↔ SOFT-AP WPA2 · CH 6

OPERATOR

Phone browser

`192.168.4.1 · no app, no router`

- Press-and-hold drive pads, PWM 60–255
- EMERGENCY STOP / RESUME, latched in firmware
- DOSE, manual probe ↓ ↑, pump disable
- Single drive owner: second client gets NAK `controller_busy`

↔ MJPEG 192.168.4.2

OPTIONAL TIERS

ESP32-CAM & laptop

`video · USB 115200 console`

- AI-Thinker CAM joins the rover AP, static IP
- Python keyboard teleop with CSV telemetry log
- Plain-text protocol: ACK / NAK per line
- `calibrate.py` prints moisture & battery constants

DOSING SEQUENCE — DRIVE FROZEN THROUGHOUT

01	02	03	04	05	06
Pre-soak pump 1.5 s	Servo eases probe down	Dwell 0.8 s, read soil	Micro-dose 1.5 s	Probe retracts, relays off	Drive released

03 ENGINEERING THAT KEEPS IT SAFE AND HONEST

Layered stop logic

- 1 s dead-man: closing the page stops the motors
- Forward blocked below 25 cm on any of 3 sonars
- Latched emergency halt, explicit RESUME only
- Drive inhibited below 9.9 V pack voltage
- ESP32 overtemp inhibit 85 °C, clears at 80 °C

Signal & power discipline

- Sonar echoes through 10k/10k dividers to 3.3 V
- Moisture sensor on 3.3 V, never 5 V
- 10k pull-up on the active-low relay input
- LM2596 buck set to 5.00 V before any board is attached
- 16× ADC oversampling, median-of-3 ultrasonic

Made to be reproduced

- Pin map identical to the full rover firmware
- Every tuning constant in one `config.h`
- Circuit guide + wiring and pinout SVGs
- Shared network header so both boards cannot drift
- One-command PlatformIO build and flash

04 HONEST STATUS & VALIDATION GATES

Working bench build, field trial pending

Firmware, mobile control page, camera firmware, laptop console, calibration tool, wiring guide and BOM are complete in `basic-bot/`. Agronomic claims are **not made**: the low-cost NPK probe is logged as a trend, not used to prescribe fertilizer, and spraying stays manual. What this poster claims is a controllable, interlocked, instrumented rover — nothing about yield.

Firmware, mobile page & console	built
Wiring guide, BOM & pin map	built
Bench test: drive, sonar, dosing	in progress
Soil-plot traversal on battery	in progress
Probe vs. lab soil-test correlation	gate